CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\mathrm{CuO + H_2 \longrightarrow Cu + H_2O}$, the substance oxidised is: 1

(a) $\mathrm{H_2O}$ **(b)** Cu **(c)** $\mathrm{H_2}$ **(d)** CuO

2. $\mathrm{CaCO_3 \longrightarrow CaO + CO_2}$ is an example of: 1

(a) electrolytic decomposition **(b)** thermal decomposition
(c) displacement **(d)** photochemical decomposition

3. When the green crystals of ferrous sulphate are heated strongly, the residue left is: 1

(a) reddish-brown **(b)** blue **(c)** white **(d)** black

4. The brown flaky substance formed on iron on long exposure to moist air is: 1

(a) iron sulphide **(b)** iron carbonate
(c) iron chloride **(d)** hydrated iron(III) oxide

5. The yellow precipitate formed when lead nitrate solution is mixed with potassium iodide is: 1

(a) PbO **(b)** KI **(c)** $\mathrm{PbI_2}$ **(d)** $\mathrm{KNO_3}$

6. The values of x, y and z in $x\,\mathrm{Pb(NO_3)_2 \longrightarrow} y\,\mathrm{PbO} + z\,\mathrm{NO_2} + \mathrm{O_2}$ are respectively: 1

(a) 2, 2, 4 **(b)** 1, 1, 2 **(c)** 2, 4, 2 **(d)** 4, 2, 2

7. Chips manufacturers flush the packet with nitrogen to: 1

(a) keep the chips hot **(b)** make them crispier by reaction
(c) make the packet heavy **(d)** prevent oxidation (rancidity) of the oil

8. Coating iron with a layer of zinc is called: 1

(a) galvanisation **(b)** electrolysis **(c)** vulcanisation **(d)** rancidity

9. The gas evolved when dilute hydrochloric acid is added to zinc granules is: 1

(a) hydrogen **(b)** carbon dioxide **(c)** chlorine **(d)** oxygen

10. The black coating formed on copper when heated in air is: 1

(a) copper carbonate **(b)** copper nitrate **(c)** copper sulphide **(d)** copper(II) oxide

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are oil and fat containing food items flushed with nitrogen? 2

12. What is a redox reaction? Name the substance oxidised in $\mathrm{CuO + H_2 \longrightarrow Cu + H_2O}$. 2

13. Write the balanced equation: barium chloride + aluminium sulphate gives barium sulphate + aluminium chloride. 2

14. Why is respiration considered an exothermic process? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Identify the substance oxidised and reduced in 3
(i) $4\,Na + O_2 \longrightarrow 2\,Na_2O$
(ii) $CuO + H_2 \longrightarrow Cu + H_2O$.

16. A shiny brown element X on heating in air becomes black. Name X and the black substance. Write the equation. 3

17. Ferrous sulphate crystals are heated. State three observations. 3

OR

Write balanced equations with state symbols for: 3
(a) hydrogen + chlorine,
(b) sodium + water.

18. Why is silver chloride stored in dark-coloured bottles? Give the equation. 3

OR

State the type of each reaction: 3
(i) $CaO + H_2O \longrightarrow Ca(OH)_2$
(ii) $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$
(iii) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Lead nitrate powder is heated in a dry boiling tube. State the observation, write the equation and name the type. Name the gases. 5

OR

Explain exothermic and endothermic reactions. Classify: burning of natural gas, respiration, decomposition of vegetable matter into compost, thermal decomposition of CaCO3. 5

20. Zinc granules are treated with dilute sulphuric acid. State the observations, write the equation, test the gas and say why the test tube becomes warm. 5

OR

What is a decomposition reaction? Give one equation each for decomposition by heat, electricity and light. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND